

Computing and Using Determinants

Property 1

$\hookrightarrow$ An invertible matrix A has $\det(A) \neq 0$
 A singular matrix A has $\det(A) = 0$

Property 2

$\hookrightarrow$ Upper or Lower Triangular
 Diagonal $\left. \vphantom{\begin{matrix} \text{Upper or Lower Triangular} \\ \text{Diagonal} \end{matrix}} \right\} \det(A) = (\text{Product of diagonal entries})$

Proof:

$$A = \begin{bmatrix} a & x & y \\ 0 & b & z \\ 0 & 0 & c \end{bmatrix}$$

$$\begin{aligned} \det(A) &= a(bc - 0) - x(0 - 0) + y(0 - 0) \\ &= abc \end{aligned}$$

$$A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

$$\begin{aligned} \det(A) &= a(bc - 0) - 0 + 0 \\ &= abc \end{aligned}$$

Property 3

$\hookrightarrow$ For a transpose matrix $\rightarrow \det(A^T) = \det(A)$

transposing a matrix only swaps rows and columns,
it does not affect the determinant.

Property 4

↳ For an elementary matrix E ,

$$\det(EA) = \det(E) \det(A)$$

Property 5

↳ For any square matrices A and B ,

$$\det(AB) = \det(A) \det(B)$$

We will be proving both property 4 and 5 here:

Proof:

Case 1:

If A (or B) is not invertible
then $\det(A) = 0$

This also implies $\text{Rank}(A) < n$ (column numbers)

Also $\text{Rank}(AB) \leq \text{Rank}(A)$ (Basic idea of linear transformation)

So, AB is not invertible either.

$$\therefore \det(AB) = 0 = \det(A) \det(B)$$

Case 2: A (also B) is invertible.

Suppose A is elementary matrix. (Explained in 2.2)

There were three cases:

Subcase 1: Row Swapping

$$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Here, $\det(E_1) = -1$; odd swap $\det(E_2) = +1$; even swap

Also we know, odd numbers of swaps will give $\det(A) = -1$
even numbers of swaps will give $\det(A) = +1$
 $\therefore \det(EA) = \det(E) \det(A)$

Subcase 2: Row Scaling

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Applying E_2 to a matrix multiplies the second row by c

Here, $\det(E) = c$.

So, depending on the scaling value $\det$ changes.

Applying E to a matrix A would scale second row by c which implies that $\det$ will also be multiplied by c .

$$\therefore \det(EA) = \det(E) \det(A)$$

Subcase 3. Row Addition $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ k & 0 & 1 \end{bmatrix} \leadsto$ Applying E_3 to a matrix adds k times the first row to the third row.

Here, $\det(E) = 1$

$\det$ will always be 1 for any row addition.

This special type is elimination matrix.

But how do we know that row addition does not change the determinant of a matrix?

The proof of this is not in our syllabus. But learn the surface idea:

⊛ $\det$ of a matrix represents the volume (or area in 2D) of the shape formed by its row vectors.

When you add a multiple of one row to another, you're shearing the shape. Shearing does not change volume.

So, the determinant remains the same.

$$\circ \det(EA) = \det(E) \det(A)$$

$$\text{So, } \det(EA) = \det(E) \det(A)$$

Now, we know that any matrix A can be written

as, $A = E_m \dots E_2 E_1$

Now $AB = E_m \dots E_2 E_1 B$

$$\Rightarrow \det(AB) = \det(E_m \dots E_2 E_1 B)$$

$$= \det(E_m) \det(\dots E_2 E_1 B) \quad \left[\because \det(EA) = \det(E) \det(A) \right]$$

$$= \det(E_m) \dots \det(E_2) \det(E_1) \det(B)$$

$$= \det(E_m \dots E_2 E_1) \det(B) \quad \left[\because \det(E) \cdot \det(A) = \det(EA) \right]$$

$$= \det(A) \det(B)$$

$$\therefore \det(AB) = \det(A) \det(B) \quad (\text{proved})$$

(You can ignore all three subcases proof and directly use $\det(EA) = \det(E) \det(A)$ in exam script)

Cramer's Rule to $Ax = b$

Cramer's Rule expresses each solution x_i as:

$$x_i = \frac{\det(B_i)}{\det(A)} ; \quad \begin{array}{l} B_i \text{ is the matrix obtained} \\ \text{by replacing the } i\text{th column} \\ \text{of } A \text{ with vector } b. \end{array}$$

Example 1:

Solve:

$$\begin{aligned} 3x_1 + 4x_2 &= 2 \\ 5x_1 + 6x_2 &= 4 \end{aligned}$$

Solⁿ:

Here, $A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$

$$\det(A) = -2$$

Now, $B_1 = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix}$

$$\det(B_1) = -4 \quad \therefore x_1 = \frac{\det(B_1)}{\det(A)} = \frac{-4}{-2} = 2$$

$$B_2 = \begin{bmatrix} 3 & 2 \\ 5 & 4 \end{bmatrix}$$

$$\det(B_2) = 2 \quad \therefore x_2 = \frac{\det(B_2)}{\det(A)} = \frac{2}{-2} = -1$$

∴ $x = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

Example 2:

$$x_1 + 2x_2 + 3x_3 = 1$$

$$4x_1 + 5x_2 + 6x_3 = 2$$

$$7x_1 + 8x_2 + 10x_3 = 3$$

Soln:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\hookrightarrow \det(A) = -3$$

$$\therefore B_1 = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 3 & 8 & 10 \end{bmatrix}$$

$$\hookrightarrow \det(B_1) = 1 \quad ; \quad \kappa_1 = \frac{\det(B_1)}{\det(A)} = -\frac{1}{3}$$

$$\therefore B_2 = \begin{bmatrix} 1 & 1 & 3 \\ 4 & 2 & 6 \\ 7 & 3 & 10 \end{bmatrix}$$

$$\hookrightarrow \det(B_2) = -2 \quad ; \quad \kappa_2 = \frac{\det(B_2)}{\det(A)} = \frac{2}{3}$$

$$\therefore B_3 = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 5 & 2 \\ 7 & 8 & 3 \end{bmatrix}$$

$$\hookrightarrow \det(B_3) = 0 \quad ; \quad \kappa_3 = \frac{\det(B_3)}{\det(A)} = 0$$

$$\sim \sqrt{\det(A)} \sim \sqrt{\det(A)} \sim \sqrt{\det(A)}$$

$$\therefore x = \begin{bmatrix} -1/3 \\ 2/3 \\ 0 \end{bmatrix}$$